HOMEWORK 2.5 · UNIT 2 · NC.8.EE.1

# Exponent Rules Review

Sep 10 / Sep 11 · Targets: EE.1-1, EE.1-2, EE.1-3, EE.1-4, EE.1-6, EE.1-7, EE.1-9, EE.1-8

Name _____ Date _____ Block _____

**Remember: Multiply** same base → **add** exponents: $x^3 \cdot x^4 = x^7$. **Divide** same base → **subtract**: $x^6 \div x^2 = x^4$. **Power to a power** → **multiply**: $(x^2)^3 = x^6$. $x^0 = 1$. $x^{-3} = \frac{1}{x^3}$. Numbers with numbers, letters with letters.

**SCORE 3.0** Approaching · the basics

Targets: EE.1-1, EE.1-2, EE.1-3, EE.1-4

**1** Find the value. $4^3$

REVIEW 2.1 EE.1-1

**2** Simplify. $x^6 \cdot x^4$

REVIEW 2.2 EE.1-2

**3** Simplify. $(2y^3)^5$

REVIEW 2.3 EE.1-2

**4** What is the value of $6^{-2}$?

REVIEW 2.3 EE.1-3

Ⓐ $\frac{1}{36}$ Ⓑ $-36$ Ⓒ $-12$ Ⓓ $36$

**5** Simplify. $\frac{30a^5b^7}{5a^4b^2}$

REVIEW 2.2 EE.1-4

**SCORE 4.0** Proficient · on target

Targets: EE.1-6, EE.1-7, EE.1-3

**6** Simplify. Use positive exponents. $(5m^{-2}n)(2m^{6}n^{-4})$

REVIEW 2.4 EE.1-6

**7** **Leo** made a mistake. Circle it. Explain it. Fix it.

REVIEW 2.4 EE.1-7

LEO'S WORK
$(3x^{2})^{3} = 9x^{6}$

**8** Which are equal to $\frac{1}{8}$?

EE.1-3

☐ $2^{-3}$ ☐ $8^{-1}$ ☐ $(2^{-1})^{3}$ ☐ $-8$
☐ $4^{-2}$

*Select all that apply.*

**SCORE 5.0** Mastery · go deeper

Targets: EE.1-9, EE.1-8

**9** Simplify $(x^{4}y^{-2})^{3} \cdot x^{-5}$. Justify each step.

EE.1-9

First _____ . Next _____ . The answer is _____ .

**10** Lily says $2^{-4}$ is a negative number. Is she right? Explain.

EE.1-8

Lily is _____ because $2^{-4}$ = _____ .